India Sector Rotation Model Trading Framework

Discretionary Strategy with Quantitative Support

Early Trend Detection for Indian Equity Markets

Report Date: 25 October 2026
Universe: 47 Active Rotation Indices + 2 Reference | NSE

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1. Executive Summary

47	14	4	5	10	16
Active Rotation Indices	Core Sectoral Indices	Signal Blocks	Macro Regime Stages	Macro Indicators	ML Ensemble Models

The India Sector Rotation Model Trading Framework (MTF) is a 4-layer quantitative system designed for discretionary implementation with quantitative signal support. It covers 47 actively rotated NSE sectoral and thematic indices, scored across momentum, trend, macro-regime alignment, and fundamental valuation dimensions.

The framework classifies India's business cycle into 5 macro stages using a 10-indicator composite (IIP, PMI Manufacturing, PMI Services, RBI Repo Rate, Yield Spread, Bank Credit Growth, CPI, FII Flows, GST Collections, INR/USD). Each sector receives a composite score from 0–100, mapped to a traffic light system: GREEN (>65) for overweight, AMBER (40–65) for neutral, RED (<40) for underweight. India-specific calibrations for RBI policy, FII flows, monsoon impact, and budget/earnings calendar are embedded throughout.

The composite sector score is calculated as: Momentum (35%) + Regime Alignment (30%) + Trend (20%) + Valuation (15%). A 16-model ML ensemble (Phase 3) sharpens return predictions via expanding-window walk-forward validation. The system targets annualised alpha >4% vs Nifty 50 with a Sharpe >0.8, max drawdown <25%, and sector turnover <1.5/month.

2. Framework Architecture

Layer	Name	Purpose	Status
1	Macro Regime Classification	Identify India business cycle phase (5 stages) using 10-indicator composite	Architecture defined
2	Sector Scoring Engine	Score all 47 sectors on momentum, trend, regime fit, valuation (0–100)	Data fetcher active
3	ML Ensemble Refinement	16-model ensemble to sharpen return predictions via walk-forward validation	Pending Phase 3
4	Portfolio Construction & Risk	Translate scores into allocations with drawdown controls and India overrides	Pending Phase 4

The framework operates on a monthly rebalancing cycle with 6 discrete steps: (1) Macro Update, (2) Signal Compute, (3) Composite Score + Traffic Light, (4) India Overrides (FII, INR, Monsoon, Calendar), (5) Discretionary Human Review, (6) Portfolio weights vs signals + rotation candidates.

3. Index Universe (49 Indices)

The rotation universe spans **47 active indices** across 3 tiers, benchmarked against 2 reference indices. All data is sourced from NSE via NSEPython.

3.1 Core Sectoral Indices (14)

Index	Sector	Description
NIFTY AUTO	Automobile	Automobile & auto ancillary sector
NIFTY BANK	Financials	12 most liquid & large banking stocks
NIFTY CONSUMER DURABLES	Consumer Durables	Consumer durables sector
NIFTY ENERGY	Energy	Energy sector — oil, gas, power
NIFTY FINANCIAL SERVICES	Financial Services	Banks + NBFCs + insurance + exchanges
NIFTY FMCG	FMCG	Fast moving consumer goods
NIFTY HEALTHCARE INDEX	Healthcare	Healthcare sector (broader than pharma)
NIFTY INFRA	Infrastructure	Infrastructure companies
NIFTY IT	Information Technology	IT & IT-enabled services
NIFTY MEDIA	Media	Media & entertainment sector
NIFTY METAL	Metals	Metal & mining sector
NIFTY OIL & GAS	Oil & Gas	Oil, gas & petroleum sector
NIFTY PHARMA	Pharmaceuticals	Pharmaceutical sector
NIFTY REALTY	Real Estate	Real estate sector

3.2 Sectoral Variant Indices (8)

Index	Sector	Description
NIFTY FINANCIAL SERVICES 25/50	Financial Services	Financial services with concentration cap
NIFTY FINANCIAL SERVICES EX-BANK	Financial Services	Financial services excluding banks
NIFTY MIDSMALL FINANCIAL SERVICES	Financial Services	Mid & small cap financial services
NIFTY MIDSMALL HEALTHCARE	Healthcare	Mid & small cap healthcare
NIFTY MIDSMALL INDIA CONSUMPTION	Consumption	Mid & small cap consumption theme
NIFTY MIDSMALL IT & TELECOM	Information Technology	Mid & small cap IT and telecom
NIFTY PRIVATE BANK	Financials	Private sector banks only
NIFTY PSU BANK	Financials	Public sector banks

3.3 Thematic Indices (25)

Index	Theme	Description
NIFTY CAPITAL MARKETS	Capital Markets	Brokers, exchanges, AMCs, depositories
NIFTY COMMODITIES	Commodities	Commodity-linked companies

NIFTY CPSE	PSU	Central public sector enterprises
NIFTY EV & NEW AGE AUTOMOTIVE	EV	Electric vehicle & new-age auto
NIFTY GROWSECTORS 15	Growth	15 high-growth sector index
NIFTY HOUSING	Real Estate	Housing finance & real estate ecosystem
NIFTY INDIA CONSUMPTION	Consumption	India consumption theme
NIFTY INDIA CORPORATE GROUP 25	Corporate Groups	Top 25 corporate group companies
NIFTY INDIA DEFENCE	Defence	Defence & aerospace sector
NIFTY INDIA DIGITAL	Digital Economy	Digital economy companies
NIFTY INDIA MANUFACTURING	Manufacturing	Manufacturing sector
NIFTY INDIA NEW AGE CONSUMPTION	Consumption	New-age consumption companies
NIFTY INDIA RAILWAYS & NEW AGE MOBILITY	Mobility	Railways & new-age mobility
NIFTY INDIA TOURISM	Tourism	Tourism, hotels, travel sector
NIFTY MNC	MNC	Multinational corporations listed in India
NIFTY MOBILITY	Mobility	Mobility sector
NIFTY MSCI INDIA DOMESTIC & WORLD HEALTHCARE	Healthcare	MSCI-linked healthcare index
NIFTY NON-CYCLICAL CONSUMER	Defensives	Non-cyclical consumer companies
NIFTY PSE	PSU	Public sector enterprises
NIFTY RURAL ECONOMY	Rural	Rural economy beneficiaries
NIFTY SERVICES SECTOR	Services	Services sector broad index
NIFTY SHARIAH 25	Shariah	Shariah-compliant top 25 stocks
NIFTY TOTAL MARKET	Broad Market	Total market index
NIFTY TRANSPORTATION & LOGISTICS	Logistics	Transportation & logistics sector
NIFTY500 MULTICAP MOMENTUM QUALITY 50	Factor	Momentum + quality factor index

4. Signal Block 1: Momentum Score (35%)

The momentum score captures price return across multiple time horizons, cross-sectionally ranked and weighted to produce a 0–100 normalised score. Longer horizons are weighted more heavily to capture sustained trends while the 1-month component provides responsiveness to recent shifts.

Component	Weight	Lookback	Description
MOM_1M	15%	21 trading days	1-month price return — recent momentum shift
MOM_3M	20%	63 trading days	3-month return — short-term trend confirmation
MOM_6M	25%	126 trading days	6-month return — medium-term trend strength
MOM_12M	30%	252 trading days	12-month return — primary trend signal
REL_STR	10%	63 trading days	Relative strength vs Nifty 500 benchmark

Formula: Raw_Momentum = (0.15 × MOM_1M) + (0.20 × MOM_3M) + (0.25 × MOM_6M) + (0.30 × MOM_12M) + (0.10 × REL_STR)

Each component is percentile-ranked cross-sectionally across the 47-index universe. The final momentum score is the weighted sum, normalised to 0–100.

Sources: *Mamais (2024), Rothe (2023)*

5. Signal Block 2: Trend & Technical Score (20%)

The trend score provides technical confirmation of momentum signals using moving averages, RSI, rate of change, and volume analysis. This block acts as a filter to prevent overweighting sectors with deteriorating technical structure.

Component	Weight	Calculation
MA_SIGNAL	35%	Binary: Price > 50-DMA > 200-DMA = 1 (bullish); 50-DMA < 200-DMA = 0 (bearish); else 0.5
RSI_14	30%	14-day RSI mapped to score: 50–70 = full, <30 or >80 = low, scaled proportionally
ROC_20	20%	20-day rate of change, cross-sectionally percentile-ranked
VOL_CONFIRM	15%	Volume confirmation: ratio of avg volume on up-days vs down-days over 20 days

Formula: Trend_Score = (0.35 × MA_SIGNAL × 100) + (0.30 × RSI_SCORE) + (0.20 × rank(ROC_20)) + (0.15 × VOL_CONFIRM × 100)

Source: *Rothe (2023)*

6. Signal Block 3: Macro-Regime Alignment (30%)

The macro-regime alignment block is the core differentiator of this framework. It classifies India's business cycle into 5 stages using a 10-indicator composite, then maps each sector's historical performance affinity to the current regime stage.

6.1 Macro Indicators (10)

#	Indicator	Bullish (+1)	Neutral (0)	Bearish (–1)
1	IIP (Index of Industrial Production)	YoY > 5%	2–5%	< 2%
2	PMI Manufacturing	> 52	49–52	< 49
3	PMI Services	> 53	50–53	< 50
4	RBI Repo Rate	Cutting cycle	On hold	Hiking cycle
5	Yield Spread (10Y–2Y)	Steepening	Flat	Inverting
6	Bank Credit Growth	YoY > 14%	10–14%	< 10%
7	CPI Inflation	< 4%	4–6%	> 6%
8	FII Flows (net monthly)	Net buyer	Mixed	Net seller
9	GST Collections	YoY > 12%	8–12%	< 8%
10	INR/USD	Appreciating	Stable	Depreciating

6.2 Regime Stage Classification

Macro_Composite = Average of 10 indicator scores (range: –1.0 to +1.0)

Stage	Composite Range	Characteristics	Favoured Sectors
Stage 1: Early Expansion	+0.2 to +0.6	RBI cutting, credit picking up, PMI rising	Banks, Auto, Infra, Realty, FinServ
Stage 2: Mid Expansion	+0.6 to +1.0	Broad-based growth, strong FII flows	Bank, Auto, IT, Infra, Metals, Realty, ConDurables
Stage 3: Late Expansion	–0.2 to +0.2	Growth peaking, inflation pressures	IT, Metals, Energy
Stage 4: Early Contraction	–0.6 to –0.2	RBI hiking, demand slowing	FMCG, Pharma
Stage 5: Deep Contraction	–1.0 to –0.6	Recession indicators, FII outflows	FMCG, Pharma (defensives)

Sources: *Garg & Sah (2025), Fidelity AART Framework, Jacobsen et al. (2009)*

7. Signal Block 4: Fundamental Valuation (15%)

The valuation block provides a mean-reversion overlay using z-scores of P/E, P/B, and dividend yield relative to their 5-year rolling averages. Cheaper sectors (negative PE/PB z-scores, positive DY z-scores) score higher.

Metric	Z-Score Calculation	Interpretation
PE Z-Score	(Current PE – 5yr avg PE) / 5yr std PE	Negative = cheap relative to history
PB Z-Score	(Current PB – 5yr avg PB) / 5yr std PB	Negative = cheap relative to history
DY Z-Score	(Current DY – 5yr avg DY) / 5yr std DY	Positive = higher yield = cheap

Formula: Valuation_Score = 100 – percentile_rank(avg(PE_ZSCORE + PB_ZSCORE – DY_ZSCORE))

Source: *Mateus et al. (2018)*

8. Composite Scoring & Traffic Light System

The four signal blocks are combined into a single composite score per index, which drives the traffic light classification and rotation decisions.

8.1 Composite Score Formula

Signal Block	Weight	Rationale
Momentum Score	35%	Strongest single predictor of near-term sector returns; captures market pricing dynamics
Regime Alignment	30%	Core differentiator; maps sector affinity to India business cycle stage
Trend Score	20%	Technical confirmation filter; prevents overweighting deteriorating sectors
Valuation Score	15%	Mean-reversion overlay; favours historically cheap sectors at margin

8.2 Traffic Light Classification

Signal	Composite Score	Action	Portfolio Implication
GREEN — BUY	> 65	**Overweight**	Active allocation above benchmark weight; primary rotation targets
AMBER — HOLD	40 – 65	**Neutral**	Market weight; monitor for upgrade/downgrade signals
RED — AVOID	< 40	**Underweight**	Below benchmark or zero allocation; exit candidates

Rotation threshold: A sector is only rotated into if the composite score differential vs the current holding exceeds 15 points. This prevents excessive turnover from marginal signal changes.

9. Regime Alignment Matrix

This matrix maps each core sector to Favour (100), Neutral (50), or Avoid (0) for each of the 5 macro stages. It encodes the historical cyclical affinity of Indian sectors to business cycle phases.

Sector Index	S1 Early Exp.	S2 Mid Exp.	S3 Late Exp.	S4 Early Cont.	S5 Deep Cont.
NIFTY AUTO	**Favour**	**Favour**	Neutral	Avoid	Avoid
NIFTY BANK	**Favour**	**Favour**	Neutral	Avoid	Avoid
NIFTY CONSUMER DURABLES	Neutral	**Favour**	Neutral	Avoid	Avoid
NIFTY ENERGY	Neutral	Neutral	**Favour**	Neutral	Neutral
NIFTY FINANCIAL SERVICES	**Favour**	**Favour**	Neutral	Avoid	Avoid
NIFTY FMCG	Avoid	Neutral	Neutral	**Favour**	**Favour**
NIFTY INFRA	**Favour**	**Favour**	Neutral	Avoid	Avoid
NIFTY IT	Neutral	**Favour**	**Favour**	Neutral	Neutral
NIFTY MEDIA	Avoid	Neutral	Neutral	Neutral	Avoid
NIFTY METAL	Neutral	**Favour**	**Favour**	Avoid	Avoid
NIFTY PHARMA	Avoid	Neutral	Neutral	**Favour**	**Favour**
NIFTY REALTY	**Favour**	**Favour**	Neutral	Avoid	Avoid

10. India-Specific Calibrations

Global sector rotation frameworks require India-specific adjustments. The following calibrations override or modulate the quantitative signals based on structural features of the Indian market.

#	Calibration	Mechanism	Impact
1	**RBI Dominates Cycle Timing**	Yield curve steepening leads RBI rate cuts by 3–6 months. The 10Y–2Y spread is the primary lead indicator.	Shifts regime classification earlier than other indicators suggest
2	**FII Flows Override**	INR depreciation + sustained FII outflows trigger a defensive override regardless of other signals.	Rotate to IT/Pharma (USD earners) during INR weakness
3	**Monsoon Variable**	IMD monsoon forecast (released June) affects rural demand: FMCG, Rural Banking, Auto (tractors).	Boost/penalise rural-linked sectors by ±10 pts on composite
4	**Budget & Earnings Calendar**	Signal weight reduced by 30% during Union Budget week, results season (Jan/Apr/Jul/Oct).	Prevents false signals from event-driven volatility
5	**Global Commodity Sensitivity**	Crude, Gold, Copper, DXY mapped to sector impact. Rising crude = bearish Oil & Gas input costs but bullish Energy producers.	Commodity-linked override on Metals, Energy, Oil & Gas

11. Backtest Architecture

The framework is validated via an expanding-window walk-forward backtest design. Three overlapping windows ensure the model generalises across different market regimes.

11.1 Walk-Forward Windows

Window	Training Period	Test Period	Regime Coverage
Window 1	Jan 2005 – Dec 2015	12 months OOS	GFC 2008, Recovery 2009–10, Taper 2013
Window 2	Jan 2013 – Dec 2020	12 months OOS	Demonetization 2016, COVID 2020
Window 3	Jan 2018 – Dec 2024	12 months OOS	COVID recovery, Rate hiking 2022–23

11.2 Validation Targets

Metric	Target	Description
IC Mean	> 0.05	Information Coefficient: average rank correlation between predicted and realised returns
IC IR	> 0.3	IC Information Ratio: consistency of IC across periods
Annualised Alpha	> 4% p.a.	Excess return vs Nifty 50 buy-and-hold benchmark
Sharpe Ratio	> 0.8	Risk-adjusted return on the rotation portfolio
Max Drawdown	< 25%	Worst peak-to-trough decline in portfolio value
Hit Rate	> 55%	Percentage of months where rotation outperforms benchmark
Turnover	< 1.5 sectors/month	Average number of sector rotations per rebalancing

Sources: *Jacobsen et al. (2009), Miao & Polak (2023), Mateus (2018)*

12. Portfolio Construction & Risk

The final portfolio allocation translates composite scores into sector weights subject to concentration limits and drawdown controls. The process is designed to support discretionary decision-making, not fully automated execution.

12.1 Monthly Dashboard Process (6 Steps)

Step	Description
Step 1: Macro Update	Update 10 macro indicators, compute composite, classify regime stage
Step 2: Signal Compute	Run all 4 signal blocks across 47 indices, produce ranked scorecard CSV
Step 3: Composite Score	Weighted sum of signal blocks, classify traffic light (Green/Amber/Red)
Step 4: India Overrides	Apply FII flow, INR trend, Monsoon, and Calendar event flags
Step 5: Discretionary Review	Human review layer — override signals based on qualitative judgment
Step 6: Portfolio Action	Compare current portfolio weights vs signals, identify rotation candidates

12.2 Risk Controls

- **Rotation threshold:** Minimum 15-point composite score differential to trigger rotation
- **Max drawdown control:** Target < 25% peak-to-trough; automatic de-risking at –20%
- **Turnover limit:** < 1.5 sector rotations per month to control transaction costs
- **Event window dampening:** 30% signal weight reduction during budget week and earnings seasons
- **Defensive override:** Automatic shift to FMCG/Pharma on sustained FII outflow + INR depreciation

13. Implementation Roadmap

Phase	Component	Deliverable	Status
Phase 1	Repository Structure & Config	index_universe.py, regime matrix, signal weights	**Complete**
Phase 2.1	Data Fetcher	OHLCV for 49 indices, quality checks, incremental refresh	**Active**
Phase 2.2	Momentum Score	5-component momentum block (1M/3M/6M/12M/REL_STR)	**Pending**
Phase 2.3	Trend Score	4-component trend block (MA/RSI/ROC/VOL)	**Pending**
Phase 2.4	Regime Alignment	10-indicator macro composite + stage classification	**Pending**
Phase 2.5	Valuation Score	PE/PB/DY z-score valuation block	**Pending**
Phase 3	Backtest Validation	Walk-forward backtest, IC analysis, 16-model ensemble	**Pending**
Phase 4	Dashboard & Output	Master Excel dashboard, automated scorecard generation	**Pending**

14. Disclaimer

This document describes a quantitative research framework for sector rotation in Indian equity markets. It is intended solely for research and discretionary investment strategy development purposes.

This is **not investment advice**. Past performance of any model, signal, or backtest does not guarantee future results. The framework has not yet completed full backtest validation (Phase 3 pending). No representation is made that any account will or is likely to achieve profits or losses similar to those discussed. Sector rotation strategies involve significant risks including but not limited to market risk, liquidity risk, and model risk.

Data sourced from NSE via NSEPython. Macro indicator thresholds are illustrative and subject to calibration based on ongoing research.

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